Escape speed

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Learning target: Explain escape in an ideal gravitational-energy model.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 4, Day 17.*
Inelastic; 3 J can become deformation, thermal energy or sound. Total energy is conserved.

02. Worked example: explain the model
Specific K 12.5 J/kg and U-12.5 J/kg; sum zero at the threshold.

03. Read and interpret the model
No; enough initial K to rise out of the gravitational well; gravity weakens but does not abruptly stop.

04. Apply the physics
10 J.

05. Explain your reasoning
Atmosphere, thrust history, fuel mass, rotation, trajectory and engineering constraints are omitted.

06. Review: connect an earlier idea *Review: Unit 4, Day 7.*
+480 J; the object gains gravitational potential energy.

07. Exit check
No; it remains bound under the assumptions.

08. STRETCH: test the model
Circular speed maintains an ideal near-surface orbit; escape threshold reaches arbitrarily far without further thrust, ignoring atmosphere.

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01. Warm-up: bring back an earlier idea *Review: Unit 4, Day 17.*
Momentum 30 kg·m/s before and after. K_i=150 J; K_f=78 J. Inelastic; 72 J leaves kinetic energy.

02. Worked example: explain the model
Specific K 12.5 J/kg and U-12.5 J/kg; sum zero at the threshold.

03. Read and interpret the model
No; enough initial K to rise out of the gravitational well; gravity weakens but does not abruptly stop.

04. Apply the physics
Circular√50=7.07 m/s; escape√100=10 m/s; ratio√2.

05. Explain your reasoning
Atmosphere, thrust history, fuel mass, rotation, trajectory and engineering constraints are omitted.

06. Review: connect an earlier idea *Review: Unit 4, Day 7.*
ΔU=-480 J; gravity does +480 J work.

07. Exit check
New escape speed is half.

08. STRETCH: test the model
Escape speed scales√(M/R); first√2 times, second unchanged.